SYNOPSIS

Time:- 2 hr **Semester:-V**

Class:-TYDS **Marks:-30**

Subject:-Data Engineering (DE)

SET A

Q1)Create an RDD using parallelize()

Code:

```
from pyspark import SparkContext
sc = SparkContext("local", "RDDExample")
numbers = [10, 20, 30, 40, 50]
rdd = sc.parallelize(numbers)
print(rdd.collect())
sc.stop()
```

Output:

```
[10, 20, 30, 40, 50]
```

Q2)Read and Write Data Using CSV Format

Code:

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("CSVExample").getOrCreate()
data = [
    (1, "Amit", 80),
    (2, "Neha", 90),
    (3, "Raj", 75)
]
df = spark.createDataFrame(data, ["ID", "Name", "Marks"])
df.write.mode("overwrite").option(
    "header", True
).csv("student_output")

result = spark.read.option(
    "header", True
).schema(df.schema).csv("student_output")

result.show()
spark.stop()
```

Output:

```
+---+----+-----+
| ID|Name|Marks|
+---+----+-----+
|  1|Amit|   80|
|  2|Neha|   90|
|  3| Raj|   75|
+---+----+-----+
```

SET B

Q1)Read a Text File into an RDD

Code:

```
# First, create the sample.txt file programmatically
with open("sample.txt", "w") as f:
    f.write("Hello Spark\nWelcome to Python\nApache Spark is easy\n")


from pyspark import SparkContext
sc = SparkContext("local", "TextExample")
rdd = sc.textFile("sample.txt")
print(rdd.collect())
sc.stop()
```

Output:

```
['Hello Spark', 'Welcome to Python', 'Apache Spark is easy']
```

Q2)Implement an Inner Join Using Spark SQL

Code:

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("InnerJoin").getOrCreate()

students = [
    (1, "Amit"),
    (2, "Neha"),
    (3, "Raj")
]
```

```
marks = [
    (1, 85),
    (2, 90),
    (4, 70)
]

df1 = spark.createDataFrame(students, ["ID", "Name"])
df2 = spark.createDataFrame(marks, ["ID", "Marks"])

result = df1.join(df2, "ID", "inner")
result.show()
spark.stop()
```

Output:

```
+---+----+-----+
| ID|Name|Marks|
+---+----+-----+
|  1|Amit|   85|
|  2|Neha|   90|
+---+----+-----+
```

SET C

Q1)Convert an RDD into a DataFrame

Code:

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("RDDToDF").getOrCreate()
data = [(1, "Amit"), (2, "Neha"), (3, "Raj")]
rdd = spark.sparkContext.parallelize(data)
df = rdd.toDF(["RollNo", "Name"])
df.show()
spark.stop()
```

Output:

```
+------+----+
|RollNo|Name|
+------+----+
|     1|Amit|
|     2|Neha|
|     3| Raj|
+------+----+
```

Q2)Implement Row Number and Ranking Using Window Functions

Code:

```
from pyspark.sql import SparkSession
from pyspark.sql.window import Window
from pyspark.sql.functions import row_number, rank

spark = SparkSession.builder.appName("Ranking").getOrCreate()

data = [
    ("Amit", 80),
    ("Neha", 90),
    ("Raj", 75),
    ("Priya", 90)
]

df = spark.createDataFrame(data, ["Name", "Marks"])

w = Window.orderBy(df.Marks.desc())

df.select(
    "Name",
    "Marks",
    row_number().over(w).alias("RowNumber"),
    rank().over(w).alias("Rank")
).show()

spark.stop()
```

Output:

```
+-----+-----+---------+----+
| Name|Marks|RowNumber|Rank|
+-----+-----+---------+----+
| Neha|   90|        1|   1|
|Priya|   90|        2|   1|
| Amit|   80|        3|   3|
|  Raj|   75|        4|   4|
+-----+-----+---------+----+
```

SET D

Q1)Filter and Sort Data in a DataFrame

Code:

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("FilterSort").getOrCreate()
data = [
    (1, "Amit", 80),
    (2, "Neha", 65),
    (3, "Raj", 90),
    (4, "Priya", 75)
]
df = spark.createDataFrame(data, ["RollNo", "Name", "Marks"])
print("Students scoring more than 70:")
df.filter(df.Marks > 70).show()

print("Students sorted by marks:")
df.orderBy("Marks").show()
spark.stop()
```

Output:

```
Students scoring more than 70:
+------+-----+-----+
|RollNo| Name|Marks|
+------+-----+-----+
|     1| Amit|   80|
|     3|  Raj|   90|
|     4|Priya|   75|
+------+-----+-----+

Students sorted by marks:
+------+-----+-----+
|RollNo| Name|Marks|
+------+-----+-----+
|     2| Neha|   65|
|     4|Priya|   75|
|     1| Amit|   80|
|     3|  Raj|   90|
+------+-----+-----+
```

Q2)Apply groupBy() and count() on DataFrames

Code:

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("GroupCount").getOrCreate()
data = [
    ("Amit", "IT"),
    ("Neha", "DS"),
    ("Raj", "IT"),
    ("Priya", "DS"),
    ("Karan", "IT")
]
df = spark.createDataFrame(data, ["Name", "Department"])
df.groupBy("Department").count().show()
spark.stop()
```

Output:

```
+----------+-----+
|Department|count|
+----------+-----+
|        DS|    2|
|        IT|    3|
+----------+-----+
```